Lists and indexing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Python readiness portfolio

THE CONNECTED WEEK

One connected list-and-index record.

Explain positions, trace an update, diagnose an invalid access and guard an empty list.

Keep one habit: Keep initial values, mutation, valid indices, length, output order and source limits together.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Loops and trace tables. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Boolean conditions. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from From a paper algorithm to Python. Give an example and a mistake that the idea helps you avoid.

Loops and trace tables

> **My remembered idea, example and correction**

Boolean conditions

> **My remembered idea, example and correction**

From a paper algorithm to Python

> **My remembered idea, example and correction**

W1 · Explain the procedure

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Independent initial lists matter. Main index 1 update:[4,7,4]→[4,9,4]; W2 index 0 update: [4,7,4]→[6,7,4], length3,index 2 still 4. Fresh sum[4,7,4]→15; []→0 and [7]→7. Pictured [6,1,8] prints 6,8,3 using indices 0,-1 and length; index 3then raises IndexError after first 6. Empty list has no final item; [0] has a real zero item. Never treat a loop's item value as its index automatically.

- For counts = [4, 7, 4], give the **length**, valid nonnegative indices and value at **index** 1. Explain how the two **item**s with value 4 differ.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Predict, trace and observe

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Independent initial lists matter. Main index 1 update:[4,7,4]→[4,9,4]; W2 index 0 update: [4,7,4]→[6,7,4], length3,index 2 still 4. Fresh sum[4,7,4]→15; []→0 and [7]→7. Pictured [6,1,8] prints 6,8,3 using indices 0,-1 and length; index 3then raises IndexError after first 6. Empty list has no final item; [0] has a real zero item. Never treat a loop's item value as its index automatically.

- Start with counts = [4, 7, 4]. Run counts[0] = 6. Write the resulting **list**, its **length** and the value at **index** 2. Predict before running.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Find the first error

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Independent initial lists matter. Main index 1 update:[4,7,4]→[4,9,4]; W2 index 0 update: [4,7,4]→[6,7,4], length3,index 2 still 4. Fresh sum[4,7,4]→15; []→0 and [7]→7. Pictured [6,1,8] prints 6,8,3 using indices 0,-1 and length; index 3then raises IndexError after first 6. Empty list has no final item; [0] has a real zero item. Never treat a loop's item value as its index automatically.

- A learner loops with for count in counts and tries to read counts[count] for counts = [4, 7, 4]. Identify the first attempted **index** and error. Explain how to print each value directly.

My reasoning and evidence

MAIN LESSON · SECTION 9

W5 · State an honest limit

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Independent initial lists matter. Main index 1 update:[4,7,4]→[4,9,4]; W2 index 0 update: [4,7,4]→[6,7,4], length3,index 2 still 4. Fresh sum[4,7,4]→15; []→0 and [7]→7. Pictured [6,1,8] prints 6,8,3 using indices 0,-1 and length; index 3then raises IndexError after first 6. Empty list has no final item; [0] has a real zero item. Never treat a loop's item value as its index automatically.

- Compare [1, 5, 2] and [5, 1, 2]. State their totals and first **item**s. Would storing only the total answer which first **item** was larger? Explain.

My reasoning and evidence

MAIN LESSON · SECTION 4

Try alone · Selected independent task

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Predict the three **output**s. If the second print uses values[3] instead, where does execution stop and which later **output** is not produced?

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Distinguish a value from an index

```
values = [4, 9, 2]
print(values[1])
print(values[-1])
print(len(values))
```

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep code and evidence

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Add a labelled **list**s and **index**ing **evidence** sheet to Python readiness portfolio. Show how you can distinguish a **list index** from the value stored at that position.

  Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

> **My project change, supporting evidence and remaining limit**

MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	

Evidence	My record
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use W2: Trace an update for **List**s and **index**ing. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Use the worked case for **List**s and **index**ing. Proposed explanation: "**Length** is the last nonnegative **index**". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Additional practice · Earlier W4 transfer

REFERENCE · SECTION 8

Additional practice · Earlier W4 transfer

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Start with readings = [3, 0, 3, 2]. Replace the second **item** with 5 using its **index**. Write a loop that adds all updated readings and prints one final total. Trace every update of the total.

  Then test the summing part alone on [] and [6].

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- **Index**es are positions, not values. A **list** of shelf counts is read by position.
- Python 3: counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).
- Label the three **index** positions. Predict all original **output**s, then name the first **output** and stopping point after using **index** 3.
- Support used: alone / hint / worked example. This record is practice **evidence**.

- Project connection: For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation.

  Use actual Python 3 results to check the **prediction** after reasoning first.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / PYTHON-BRIDGE/7

Lists and indexing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Draw positions

Indexes 0,1,2 contain 6,1,8 respectively.

2. Read two positions

Index 0 returns 6; index -1 returns the final value 8.

3. Check length boundary

Length is 3, but index 3 is beyond the final valid positive index 2.

CHANGE ONE THING

Replace the second print with print(counts[3]).

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

▼ Reference · Project portfolio milestone

REFERENCE · SECTION 13

Reference · Project portfolio milestone

> Keep the stated task, its own code/values and support record together.

Task and evidence record

- Project: Python readiness portfolio
- Current milestone: Python readiness portfolio. Keep runnable code, **prediction**s, actual **output**s, one repaired error and an explanation of a function on changed **input**s.
- For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 14

Extra practice · Predict, check and repair

> Use only after an attempt; keep the reserved assessment separate.

First question and hint

- **Index**es are positions, not values. A **list** of shelf counts is read by position.
- Which labelled step matches this detail? **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2.
- 1. Read two positions
- 2. Check **length** boundary
- 3. Draw positions
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Check **length** boundary **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2. **List** positions begin at zero.

A negative **index** counts from the end; the **list length** counts entries and is not itself a valid positive **index** for the final entry.

Changed-case question and hint

- Changed-case practice: Replace the second print with print(counts[3]). Which conclusion is supported by the supplied facts?
- 1. The first print shows 6, then **IndexError** stops the script before the **length** print.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: The first print shows 6, then **IndexError** stops the script before the **length** print. The attempted position does not exist.

Repair and keep the record

Review Loops and trace tables (python-bridge/6). Goal: Trace variable values before and after each loop iteration. Short practice: **List** the four visited values and their running totals.

Explain why 5 is absent and what prints for range(1,1).. Return to this week and explain what you changed.

Review Boolean conditions (python-bridge/4). Goal: Translate stated requirements into comparisons. Short practice: Give both comparison results for ages 11, 15 and 16.

Which comparison rejects each outside value, and why does and matter?. Return to this week and explain what you changed.

Review From a paper algorithm to Python (python-bridge/1). Goal: Translate a short paper calculation into a Python script. Short practice: Match each paper quantity to its Python name.

Why does the spare count get added once, and what prints when spare is five?. Return to this week and explain what you changed.

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual **output** or error, change one **input**, then run again.

Keep code, **prediction**s, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Predict, run and debug

Keep your completed work

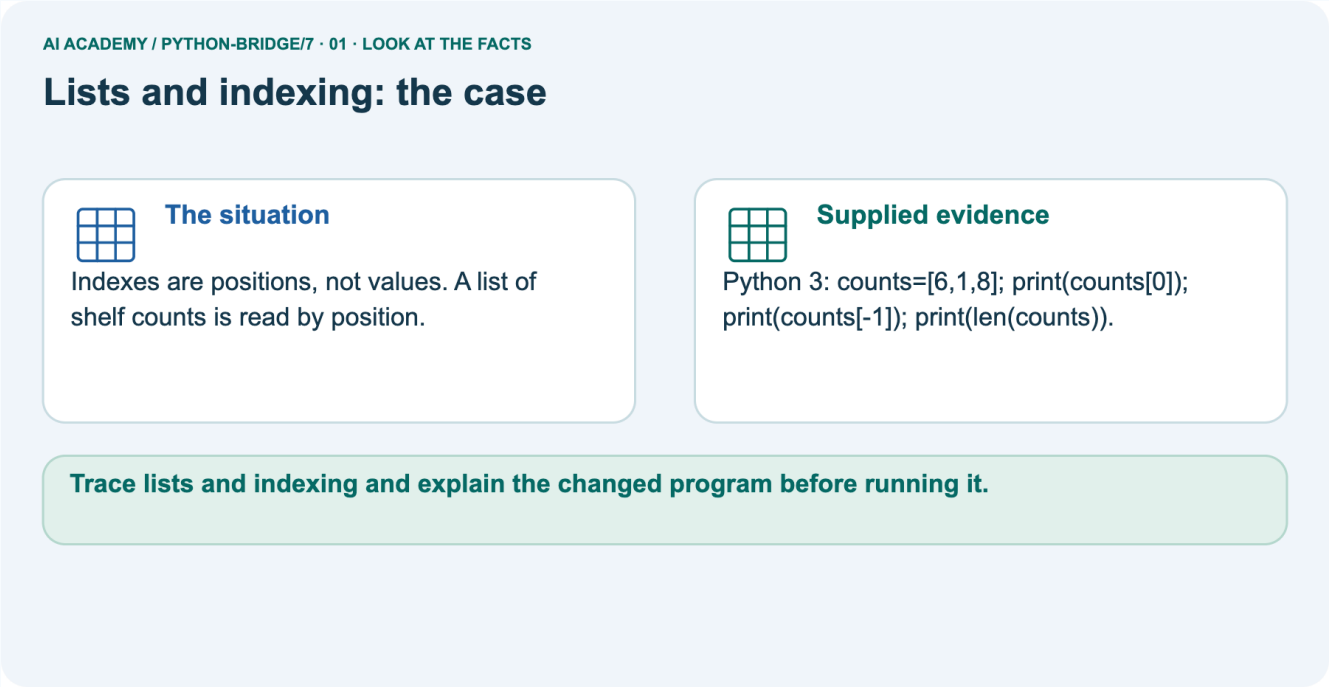
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Use the picture relevant to a current gap.

01. Lists and indexing: the case



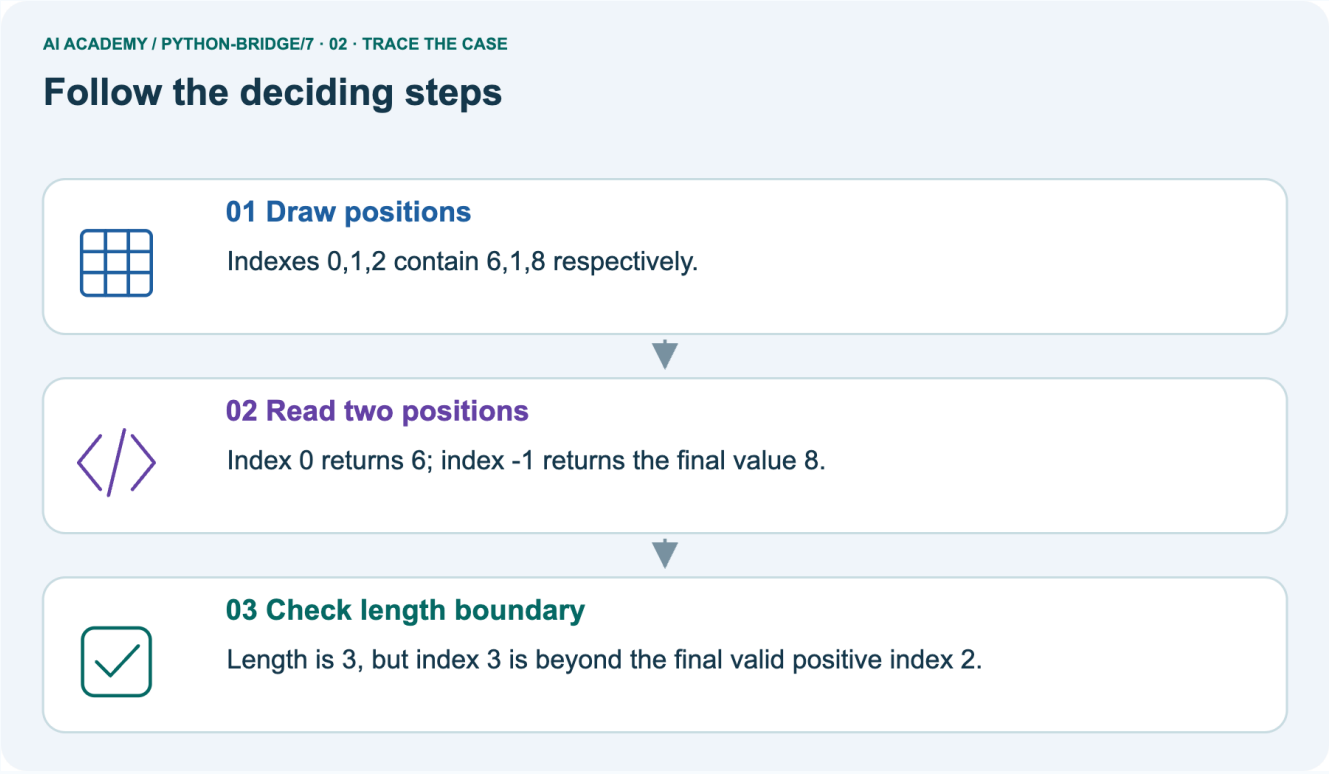
Follow the labelled picture with the matching explanation. Open the vector diagram

Indexes are positions, not values. A **list** of shelf counts is read by position. Supplied facts: Python 3: counts=[6,1,8]; print(counts[0]); print(counts[-1]); print(len(counts)).

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Draw positions: **Index**es 0,1,2 contain 6,1,8 respectively. → Read two positions: **Index** 0 returns 6; **index** -1 returns the final value 8. → Check **length** boundary: **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Zero-based list positions



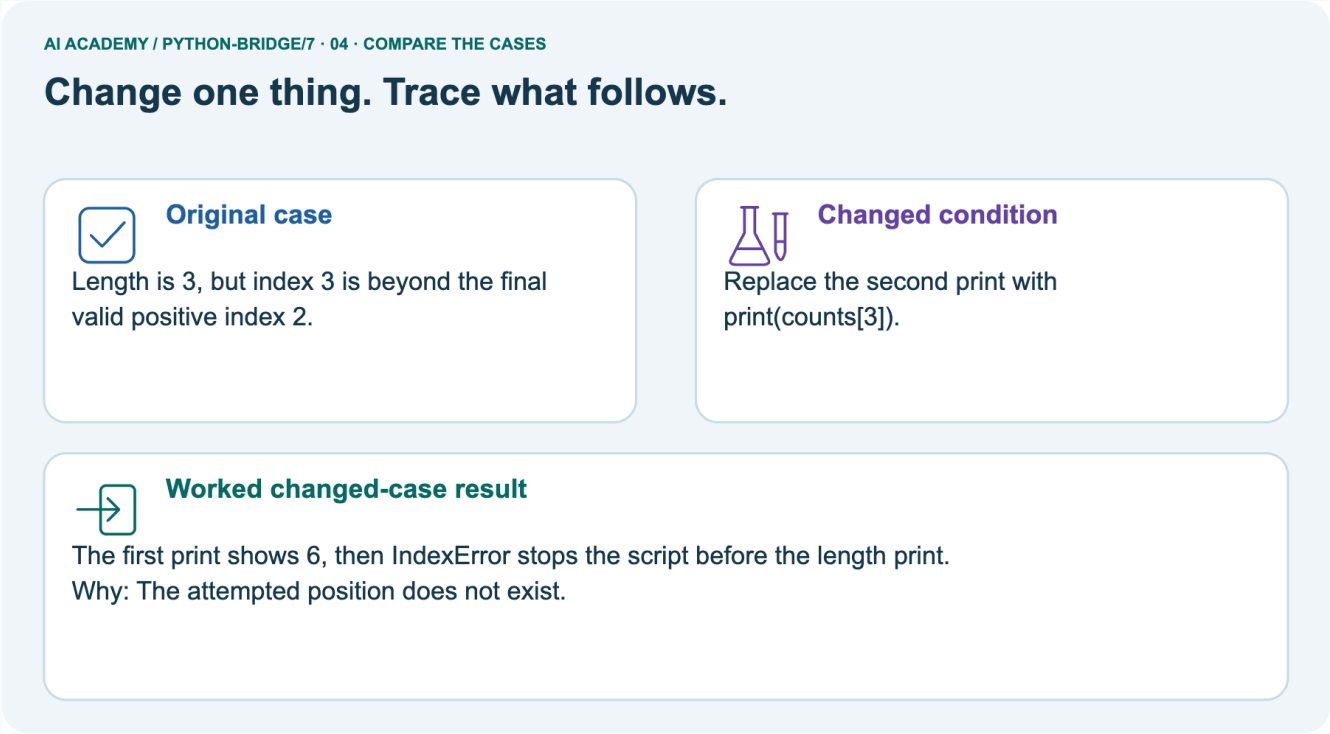
Follow the labelled picture with the matching explanation. Open the vector diagram

Zero-based **list** positions. Stored counts: **Index** 0 = 6, **Index** 1 = 1, **Index** 2 = 8.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

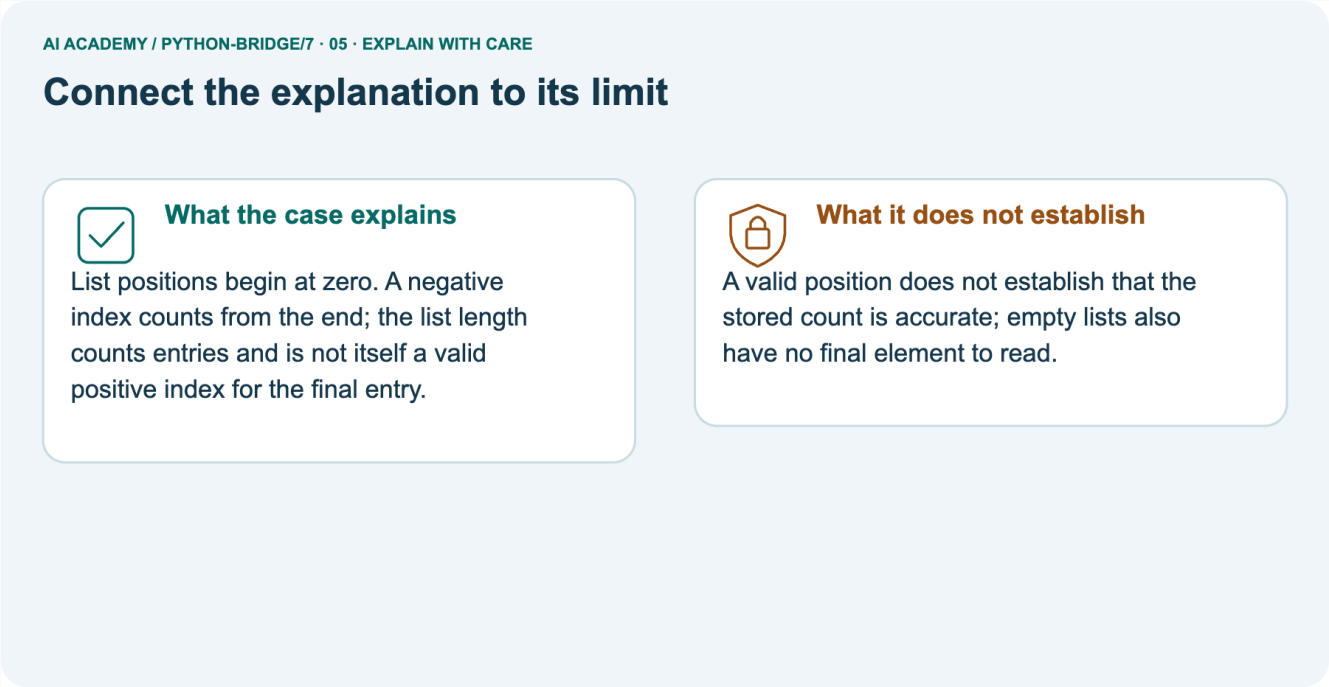
Original case: **Length** is 3, but **index** 3 is beyond the final valid positive **index** 2. Changed condition: Replace the second print with print(counts[3]).

Expected result: The first print shows 6, then **IndexError** stops the script before the **length** print. Reason: The attempted position does not exist.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

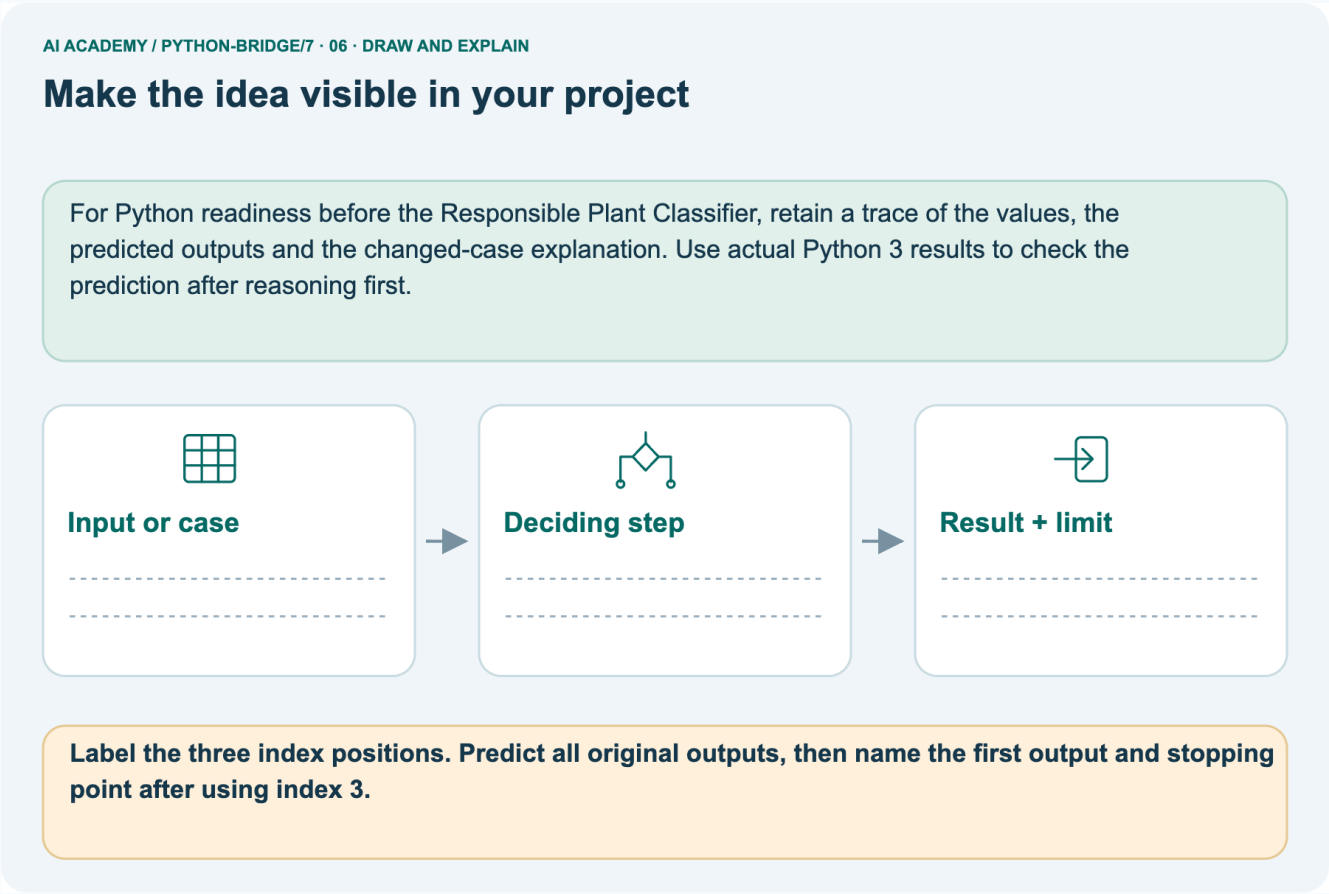
List positions begin at zero. A negative **index** counts from the end; the **list length** counts entries and is not itself a valid positive **index** for the final entry.

Limit: A valid position does not establish that the stored count is accurate; empty **list**s also have no final element to read.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For Python readiness before the Responsible Plant Classifier, retain a trace of the values, the predicted **output**s and the changed-case explanation. Use actual Python 3 results to check the **prediction** after reasoning first.

Draw your **input**, deciding step, result and limit. Supported question: Label the three **index** positions. Predict all original **output**s, then name the first **output** and stopping point after using **index** 3.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Lists and indexing: the case

02. Follow the deciding steps

03. Zero-based list positions

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.